

Installation and Operating Manual Inverter

11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3



PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3

Thank you for purchasing a PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3.

In this instruction, we have compiled all information that is important for installation and operation.

If you still have any trouble, please call our service hotline.

Service hotline

You can reach the PLATINUM service hotline as follows:

Tel +49 7522 73 777

Fax +49 7522 73 737

E-Mail service.platinum@diehlako.com

You can find more service documents at www.diehl.com/photovoltaics.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3

Menu structure

Main Menu

— Operation display	see p. 47
— Current	see p. 48
— Settings	see p. 51
— Time settings	see p. 51
— Service Menu 	see p. 65
— <i>Event list</i>	see p. 66
— <i>Parameters</i>	see p. 66
— <i>Reconfiguration</i>	see p. 68
— <i>Grid support</i>	see p. 70
— <i>Power reduction</i>	see p. 72
— <i>Startup</i>	see p. 68
— <i>Meter</i>	see p. 69
— Language	see p. 54
— Alarm volume	see p. 55
— Alarm contact	see p. 55
— LCD	see p. 56
— Feed payback	see p. 56
— System	see p. 57
— Energy meter	see p. 57
— Information	see p. 58
— Operating data	see p. 58
— System data	see p. 59
— Inverter type	see p. 60
— Inverter version	see p. 61

 The service menu is exclusively relevant for service purposes; it is not accessible directly from the menu.
For access to service menu see 8.1 Service-Menü.

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Symbols

1 Symbols

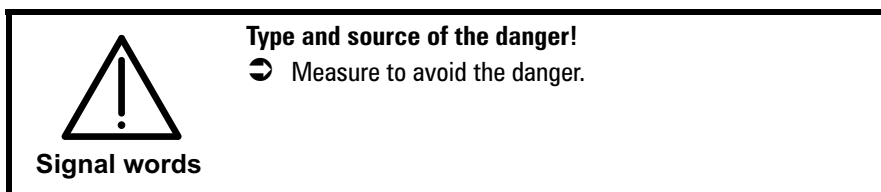
1.1 Warning notices

Classification of warning notices

The warning notices differentiate between three types of dangers indicated by the following signal words:

- ➔ **Caution** warns of material damages.
- ➔ **Warning** warns of bodily injuries.
- ➔ **Danger** warns of a danger to life.

Layout of the warning notices



1.2 Other symbols

Instructions

Layout of instructions:

- ➡ Instruction to do something.

Result of the action, if necessary.

Lists

Layout of bulleted lists:

- ➔ List level 1
 - ➔ List level 2

Layout of numbered lists:

1. List level 1
2. List level 1
 - 2.1 List level 2
 - 2.2 List level 2

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Symbols

1.3 Symbols on the product

	Vorsicht: Gehäusetemperatur >60°C !
	Caution: The temperature of the enclosure can be higher than 60°C !
	Warnung: Eingriffe in und am Gerät sind nur durch Elektro-Fachkräfte durchzuführen !
	Warning: All work inside and around the device must be done by skilled personnel only !
	Warnung: Entladezeit der Kondensatoren bis zu 30 min. !
	Warning: The capacitors within the device require up to 30 minutes to discharge !

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Safety and Dangers

2 Safety and Dangers

- Prior to starting up the inverter, read all the installation and operating instructions.
- Use inverters according to their intended use.
- Start up inverter exclusively if entire PV system complies with the national laws and safety regulations.
- Use inverters in original and technically fully intact condition without unauthorized modifications.
- Ensure that exclusively qualified electrician carry out the assembly and installation work described in this installation and operating manual.
- Ensure that inverters are installed and serviced by qualified specialists only.
- Ensure that the following is observed when mounting and connecting the inverter and the PV system:
 - ➔ directives and rules applicable in the respective country
 - ➔ regulations stipulated by trade associations and inspection authorities and agencies
 - ➔ technical connection conditions (TAB) of the relevant energy provider
 - ➔ national and international rules and regulations
- Mount inverters exclusively in specified position.
- Ensure that all protection devices are fully operational.
- Ensure that ventilation fins are neither covered nor dirty.
- Prior to installation and maintenance work, ensure that the inverter is de-energized.
- Use exclusively original spare parts.
- Operate inverters exclusively with original accessories.
- Observe conditions of use (see 13 Technical Data).

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Intended Use

3 Intended Use

- Use the inverter exclusively to feed photovoltaically converted solar energy into the public 230 V/50 Hz mains.
- Do not use the inverter in autonomous power systems.
- Do not use the inverter in vehicles.
- Do not use inverters in explosive atmospheres (flour dust, sawdust etc.).
- Do not use inverters in atmospheres where the ammonia concentration of the air exceeds 20 ppm.
- Observe warranty conditions.

Note

If these warranty conditions and the information contained in this installation and operating manual are not observed, all warranty claims will expire.

4 Function

The 3-phase transformerless inverter PLATINUM TL3 is constructed in such a way that it is not necessary to open the device for mounting or connection. All electrical connections are established exclusively with lockable plugs.

The 3-phase inverter PLATINUM TL3 is equipped with an integrated DC switch according to EN 60947-3.

A backlit graphic display shows the progress of the feed power and other data, even when the inverter is not feeding in.

Due to the degree of protection IP65 design, outdoor mounting of the 3-phase inverter PLATINUM TL3 is possible almost without limits.

4.1 Variations

The PLATINUM line includes the following variants with different power ranges (see 13 Technical Data):

- ➔ PLATINUM 11000 TL3
- ➔ PLATINUM 13000 TL3
- ➔ PLATINUM 17000 TL3
- ➔ PLATINUM 20000 TL3

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Function

4.2 Options

Upgrades for an inverter or an inverter system:

- ➔ Inverter networking via EIA 485 bus.
Much easier and more comprehensive input and monitoring options.
- ➔ Visual and acoustic indicator devices.
- ➔ Remote monitoring or remote readout with PLATINUM WebMaster
- ➔ Central display of systems with PLATINUM ViewMaster
- ➔ Evaluation of system data with PLATINUM PV-Monitor
- ➔ Remote control of AC power in connection with PLATINUM PowerCommander, PLATINUM WebMaster and a control device on the mains supply side.

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Assembly

5 Assembly

5.1 Scope of supply

- ➔ PLATINUM inverter with protector
- ➔ Mounting fixture
- ➔ 5-pole contact insert for power supply plug
- ➔ Casing for power supply plug
- ➔ Brief instruction
- ➔ 3 x plain washers, shape B M8 DIN 125-8
- ➔ 2 x pan head screws M5x20 for fixating inverter in mounting fixture

Note

A detailed installation and operating manual is available in the download area of our website www.diehl.com/photovoltaics.

The casing allows sealing according to Guida Connessioni, Edition 1.1. Fixate seal through opening on housing and hole in screw intended for this purpose (below screw head of casing).

5.2 Transport

- ➞ Transport inverter exclusively as follows:
 - ➔ clean
 - ➔ dry
 - ➔ in original packaging if possible
- ➞ Observe transport temperature of -25 °C to 70 °C.
- ➞ Ensure that temperature variations do not exceed 20 K/h.

5.3 Storage

- ➞ Store inverter exclusively as follows:
 - ➔ clean
 - ➔ dry
 - ➔ in original packaging if possible
- ➞ Observe transport temperature of -25 °C to 70 °C.
- ➞ Ensure that temperature variations do not exceed 20 K/h.

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5.4 Installation site

- Protect inverters from direct sunlight.
- Always mount inverters in a vertical position.

5.5 Unpacking

The inverter is packaged upside down. Thus, the device's bottom side (device connections) is visible when opening the packaging.

Note

When unpacking the device, the packaging grid remains latched to the device housing and serves as a support when putting the appliance on the floor. This prevents damage to the cover.

- Take inverter out of packaging by lifting it on the handles on the sides.

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5.6 Structure

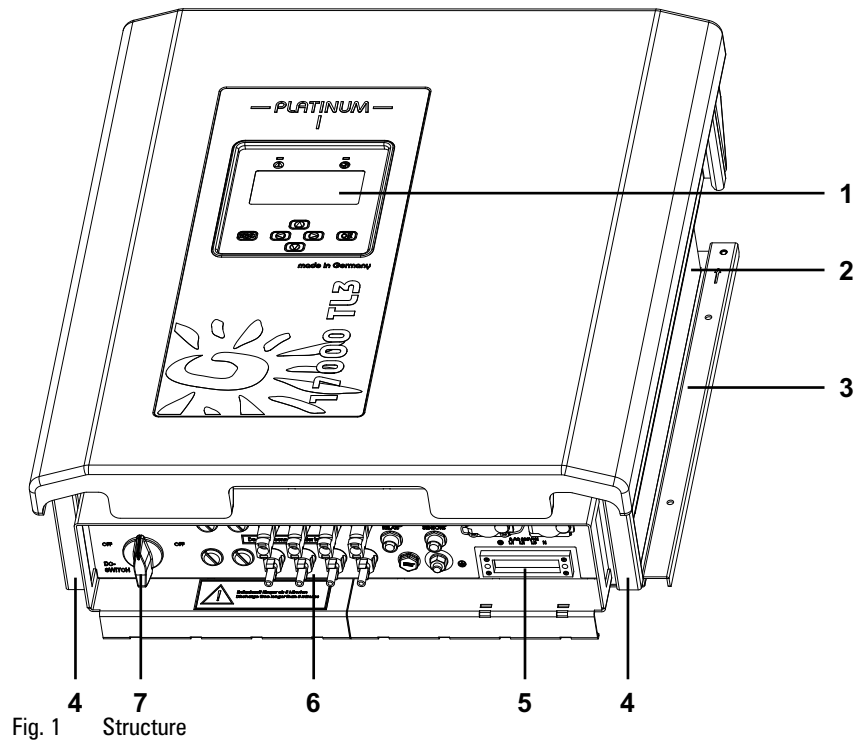


Fig. 1 Structure

- (1) Control unit
- (2) Cooling elements
- (3) Mounting fixture
- (4) Mounting screw
- (5) AC voltage connection
- (6) DC sockets
- (7) DC switch

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5.7 Mounting the inverter



Danger

Danger to life due to electric shock!

- Have the inverter opened exclusively by the PLATINUM service or service partners authorized by DIEHL Controls.



Danger

Danger to life due to incorrect connection of inverter!

- Have inverters installed by qualified specialists exclusively.
The qualified specialist personnel require a license from the relevant energy provider.



Danger

Danger to life due to inverter dropping from the wall!

- Ensure that bearing capacity of mounting wall is sufficient for weight of device (see 13 Technical Data).
- Use mounting elements appropriate for the mounting wall and the weight of the inverter unit.
- Wear protective footwear when mounting and dismantling inverters.



Danger

Risk of explosion due to unsuitable installation site!

- Do not install inverters in explosive atmospheres (flour dust, sawdust etc.).



Caution

Material damage due to unsuitable installation site!

- Observe admissible ambient temperature (see 13 Technical Data).
- Ensure that the ammonia concentration in the air is 20 ppm at most.
- Ensure that it is not possible for animals to nest in or on the inverter.
- Ensure that it is not possible for plants to grow on or into the inverter.

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Caution

Material damage due to excessive dust formation!

- ➔ Degree of protection IP65 does not apply to communication interface.
- ➔ Avoid excessive dust formation.
- ➔ Avoid dust formation with electrically conductive dust particles.

To guarantee IP65 protection rating:

- ➔ Match connecting plugs with connecting leads.
- ➔ Equip all unused connectors with dummy plugs.

Note

Because of the potential noise emission, DIEHL Controls recommends not installing the inverter in living spaces.

Protector

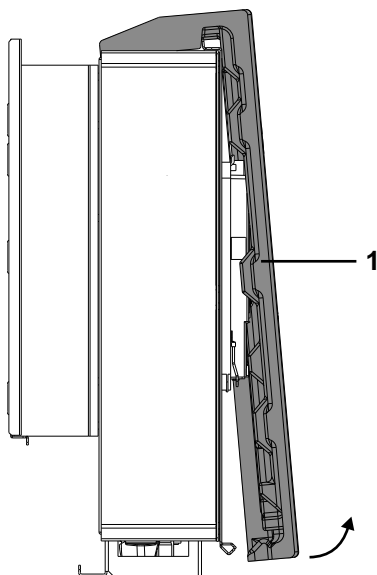


Fig. 2 Removing protector

(1) Protector

- ➔ Hold protector on underside and pull away from inverter.

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Mounting fixture

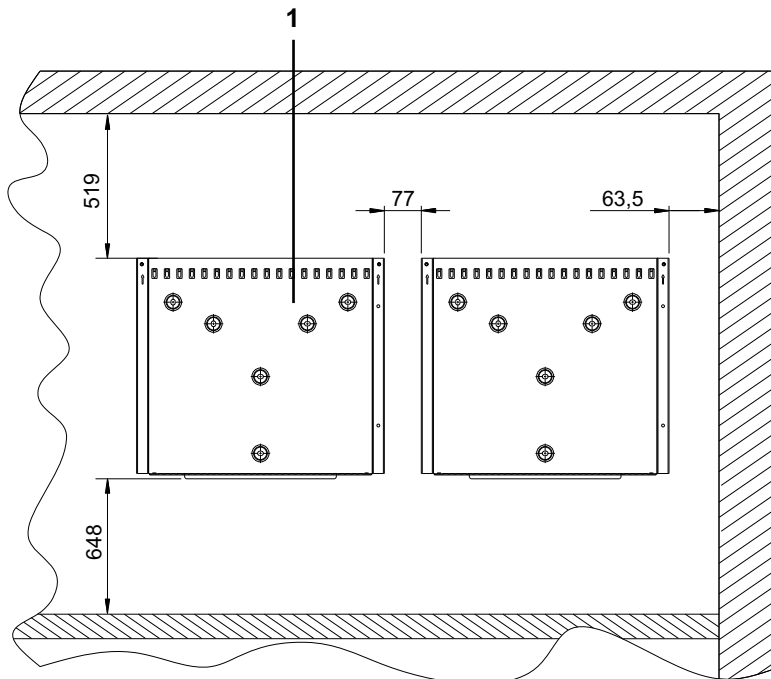


Fig. 3 Spacing for position of mounting fixture (indications in mm)

(1) Mounting fixture

Install the mounting fixture as follows:

- ➡ Unscrew mounting fixture **1** from the back of the inverter.
 - ➔ Two locking screws secure the inverter in the mounting fixture.
- ➡ Mark mounting holes for mounting fixture.
 - ➔ Observe dimensions and distances.
 - ➔ Observe a minimum distance of 50 cm to the floor.
- ➡ Drill the mounting holes and insert the screw anchors.
- ➡ Bolt down mounting fixture **1**.

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Inverter

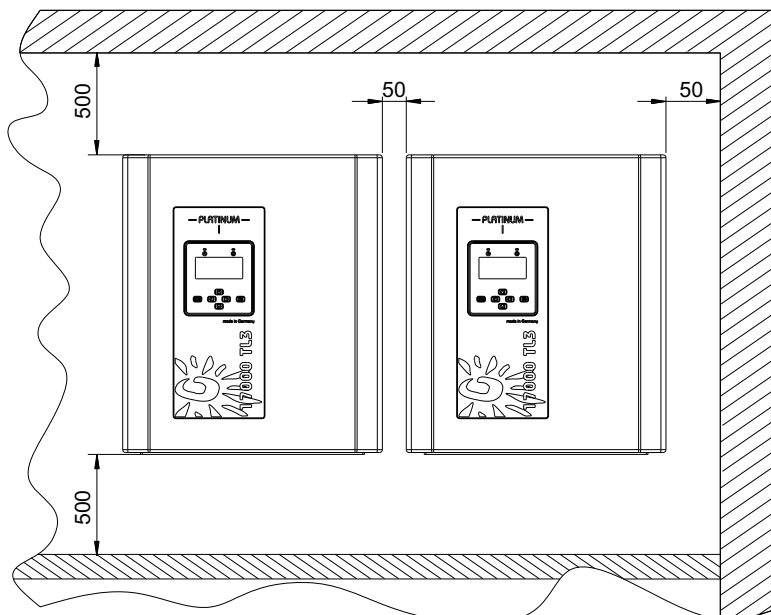


Fig. 4 Spacing for mounting of inverters (indications in mm)

- ➡ Observe a minimum distance of 50 cm to the floor.

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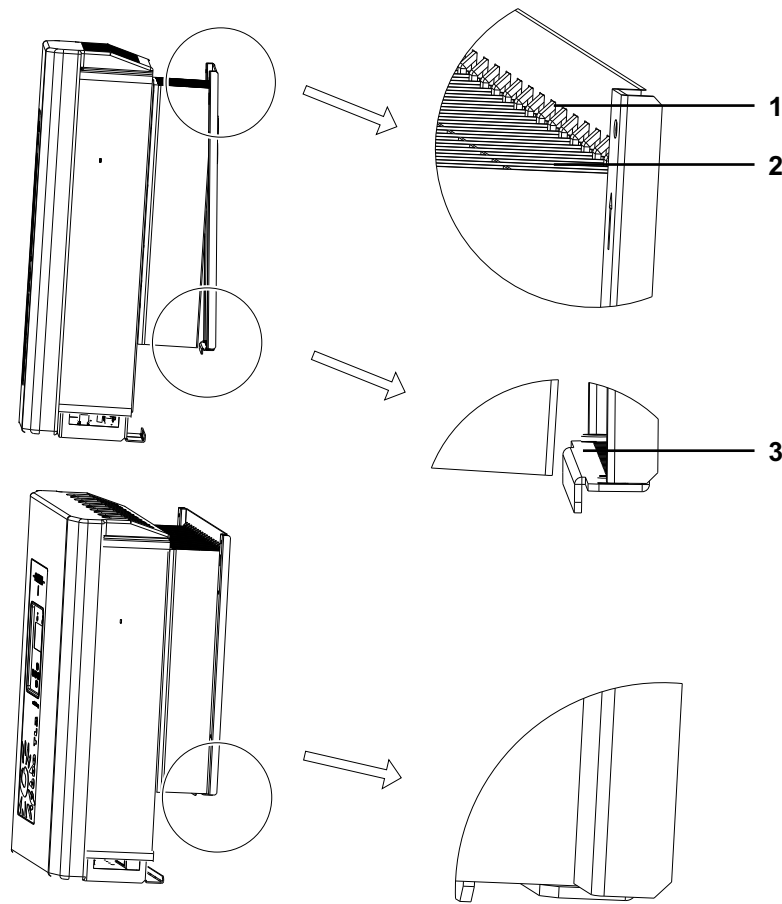


Fig. 5 Mounting the inverter

- (1) upper hook row of mounting fixture
- (2) cooling elements of inverter
- (3) lower attachment seam

Mount inverters as follows:

- Push upper end of cooling element of inverter **2** into upper hook row of mounting fixture **1**.
- Push inverter against cooling element in direction of wall and lower onto lower attachment seam **3**.
- Fixate inverter on mounting fixture from below with mounting screws M5x20.
- Verify that minimum distances are kept (see Fig. 4).
- Verify that the inverter is hanging in the mounting fixture correctly.

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5.8 Connection of the inverter



Danger

Danger to life due to high voltage!

- ➡ Observe the following order for the electronic connection:
 1. Earthing
 2. AC voltage connection
 3. DC voltage connection
 4. Connection in PLATINUM network (EIA 485)



Warning

Risk of burns due to hot surfaces!

After a longer operation with high power, housing temperatures of $> 60\text{ }^{\circ}\text{C}$ are possible.

- ➡ Do not touch hot surfaces.



Caution

Lack of performance and functionality due to inappropriate cables!

- ➡ Ensure that wire cross section areas and fuses conform to VDE 100 Part 430.
- ➡ Ensure that any network cables between two inverters are not longer than 30 m.
- ➡ Select a cross section area of DC leads and DC plug connectors that ensures minimum power loss (approx. 1%).
- ➡ Select a cross section area of AC leads that ensures minimum power loss (approx. 1%).



Caution

Material damage due to excessive voltage!

- ➡ Ensure that the max. DC voltage is not exceeded (see 13 Technical Data).
- ➡ Connect exclusively safety extra low voltage to the contact of the external indicator.

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Terminal side

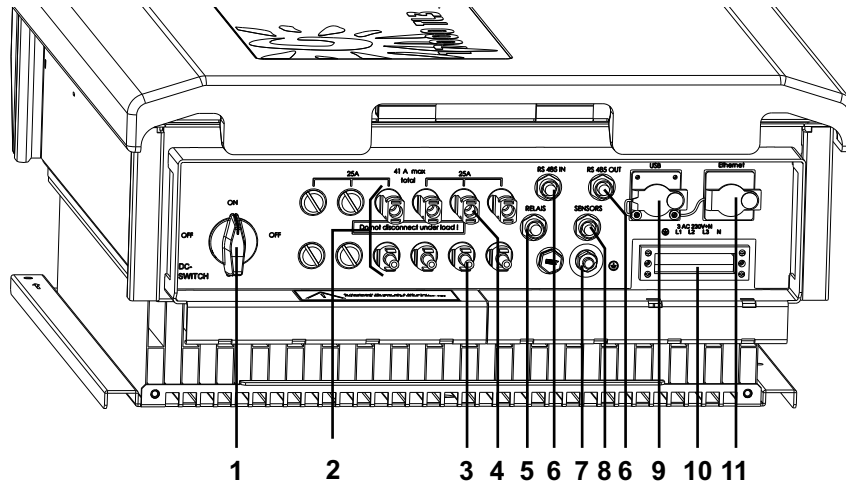


Fig. 6 Terminal side inverter

- (1) DC switch
- (2) four or six pairs of PV generator connectors
- (3) DC terminals -
- (4) DC terminals +
- (5) Relay connector (no function)
- (6) RS485 connectors (no function)
- (7) Earthing
- (8) Sensor connector (no function)
- (9) USB connectors (exclusively for service)
- (10) AC voltage connection
- (11) Ethernet connector (exclusively for service)

PLATINUM 11000 TL3 and 13000 TL3 are equipped with four pairs of PV generator connectors. PLATINUM 17000 TL3 and 20000 TL3 are equipped with six pairs of PV generator connectors.

Note

- ➡ Cover all unused connectors with provided protective covers.

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Earthing

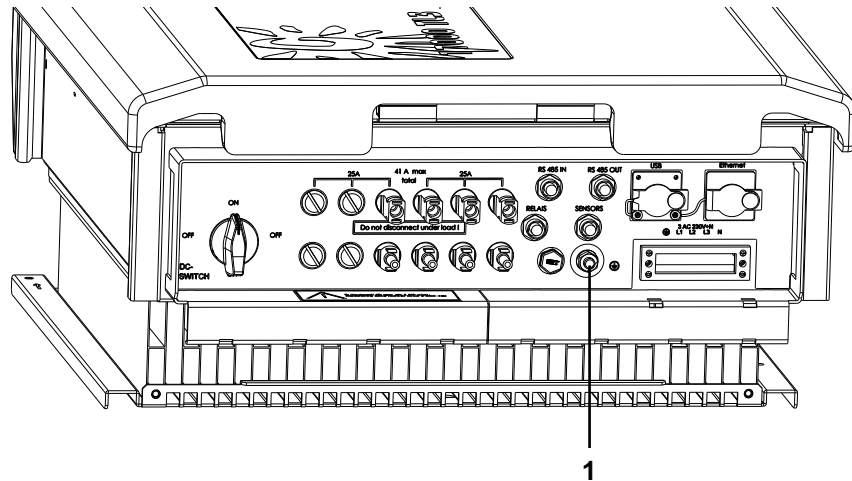


Fig. 7 Connector for earthing

(1) Connector for earthing

To ensure an optimal overvoltage protection, establish additional earthing as follows:

- Connect additional earthing to earthing connector 1.
- Select a cross section area for the earth lead that is larger than that of the mains lead (at least 10 mm²).
- Install earth lead as far as possible from mains lead and not parallel to mains lead.

FI protection

For socket circuits with up to 20 A for interior rooms, or 32 A outdoors, that are used by non-experts in the field of electrical engineering, protection by FI protection switches is required.

Inverter meets requirements regarding leakage protection according to DIN VDE 0100-712, IEC 60364-7-712:2002 and CEI 64-8/7.

Note

It is possible to operate inverters without any limitation of function or protection with Type A FI circuit breaker.

- Ensure that nominal current is at least 100 mA oer inverter.

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AC voltage connection



Danger

Danger to life due to high AC voltage!

Before connecting the inverter:

- ➞ Switch off voltage supply.
- ➞ Make sure that inverter is de-energized.
- ➞ Secure circuit breaker against switching on.
- ➞ Establish earthing connection with marked earthing connector.



Danger

Danger to life due to high AC voltage!

- ➞ Maximum permissible protection on alternating voltage side: 32 A.



Caution

Danger of electric shock and fire due to high leakage current!

Before connecting inverter to supply circuit:

- ➞ Ensure that earthing connection is established with marked earthing connector.

- ➞ Observe the following standards:
 - ➔ DIN VDE 0298-4 Application of cables and cords and current-carrying capacity
 - ➔ DIN VDE 0100; Teil 430 Protection measures for protection of cables and conductors in terms of overcurrent
 - ➔ DIN VDE 0100; part 410 Protective measures and protection against electric shock
- ➞ Use lines matching the type-dependent AC.
- ➞ Fuse lines accordingly.
 - ➔ Max. device current per phase: 25 A
- ➞ Equip leads with appropriate cable protection.
- ➞ Use exclusively provided power supply plugs.

For easier processing, DIEHL Controls recommends using fine-wire leads as lead-in for all lead diameters.

Note

Due to the threaded cable connection, the provided standard plug casings provided allow connecting a 5 x 6 mm² cable. The maximum outer diameter is limited to 18 mm. A larger plug casing is optionally available.

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The following table indicates the lead lengths depending on the conductor diameter (voltage dip = 1%).

Cross section area of lead	max. lead length
4.0 mm ²	20 m
6.0 mm ²	30 m
10.0 mm ²	50 m

To connect the cables:

- ➡ Open plug casing.
- ➡ Connect provided power supply plugs to AC lead (see Fig. 8).

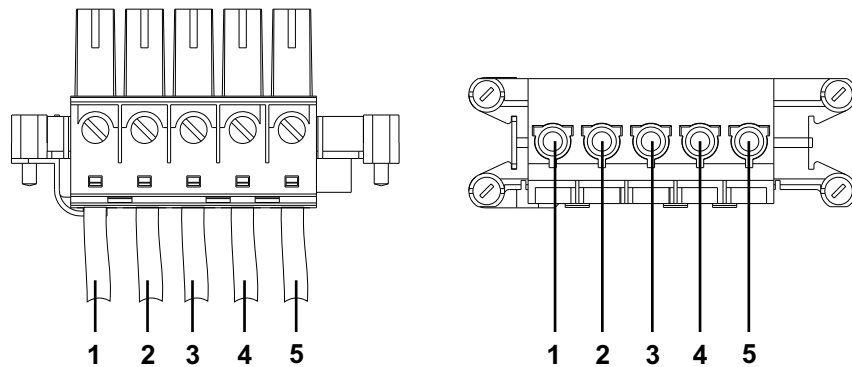


Fig. 8 Connecting provided power supply plug to AC lead

- (1) PE
- (2) L1
- (3) L2
- (4) L3
- (5) N

- ➡ Plug power supply plug into socket of inverter and fasten with screws.

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DC voltage connection



Danger

Danger to life due to high DC voltage!

- ➔ Prior to connecting the inverter, verify that voltage is present at the generator-side DC port.
- ➔ Prior to connecting the inverter, check if DC voltage polarity is correct.
- ➔ Wear insulating protective clothing and face protection if DC input.
- ➔ Remove DC cable exclusively if inverter is out of operation.



Caution

Material damage due to earth leakage!

Before connecting to inverter:

- ➔ Check PV strings for earth leakage.
- ➔ Eliminate earth leakage.
- ➔ Ensure that positive or negative terminal of PV generator are not earthed.



Caution

Material damage due to DC plug connectors that don't fit!

- ➔ Use exclusively DC cable connectors and DC sockets of the same type and manufacturer on the inverter.
- ➔ Use exclusively Multicontact DC plug connectors.
- ➔ Ensure that cable connector is correctly latched in socket.

The PV generator is connected with the Multi-Contact DC plugs and DC sockets built into the housing.

Use exclusively original Multi-Contact plugs and sockets.

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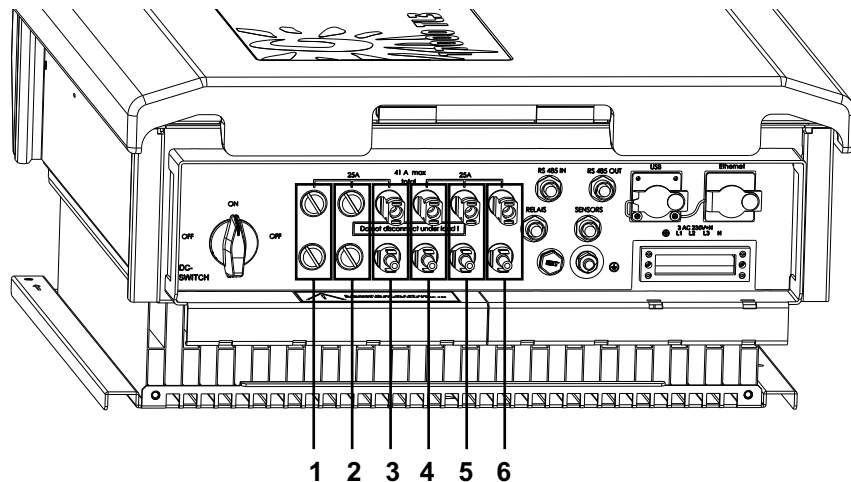


Fig. 9 DC connection

- (1) Socket pair 6
- (2) Socket pair 5
- (3) Socket pair 4
- (4) Socket pair 3
- (5) Socket pair 2
- (6) Socket pair 1

- Use exclusively leads with appropriate cross section areas.
 - ➔ Max. admissible cross section area of lead: 6 mm².
- Observe max. admissible DC current.
- Distribute DC current to socket pairs:
 - ➔ Max. DC current per DC socket pair: 25 A
 - ➔ Max. DC current of all socket pairs: 36 A

Possible distribution of connecting leads for PLATINUM 11000 TL3 and 13000 TL3:

- ➔ 2 PV connecting leads
 - ➔ socket pairs 1 and 3
 - or
 - ➔ socket pairs 2 and 4
- ➔ 3 PV connecting leads
 - ➔ socket pairs 1 2 and 3
 - or
 - ➔ socket pairs 1 3 and 4
 - or
 - ➔ socket pairs 2 3 and 4
- ➔ 4 PV connecting leads
 - ➔ socket pairs 1, 2, 3, 4

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Possible distribution of connecting leads for PLATINUM 17000 TL3 and PLATINUM 20000 TL3:

- ➔ 3 PV connecting leads
 - ➔ socket pairs 1 3 and 5
 - or
 - ➔ socket pairs 2 4 and 6
- ➔ 4 PV connecting leads
 - ➔ socket pairs 1 3, 5 and 6
 - or
 - ➔ socket pairs 2 4, 5 and 6

Establish DC voltage connection as follows:

- ➔ Establish a connection to the solar panel on the DC input with a cable with a DC plug connector.
- ➔ If more than one string is connected, ensure that the number and type of solar modules and the PV power are identical for every string.
- ➔ Put protective caps on unused plug connectors.

Connection to PLATINUM network (EIA 485)

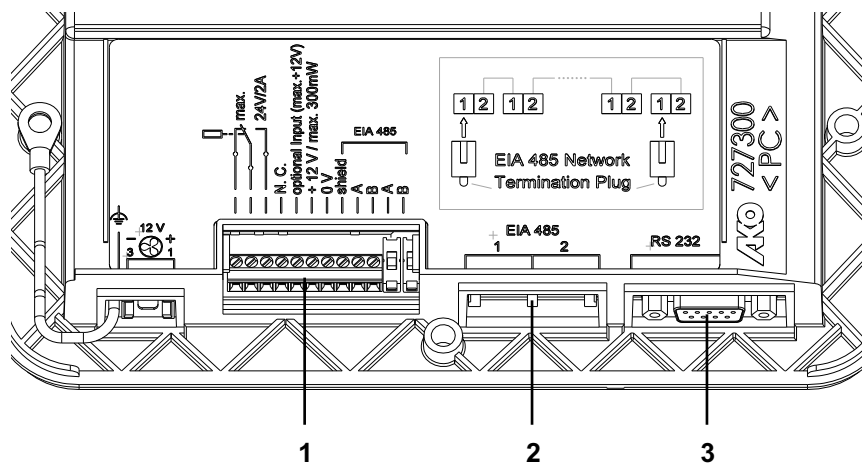


Fig. 10 Layout of network terminals

- (1) Terminal strip
- (2) EIA485 network terminals
- (3) PC interface to EIA232 (exclusively for service purposes)

It is possible to connect up to 50 PLATINUM inverters.

Note

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Assembly

Terminal strip connectors

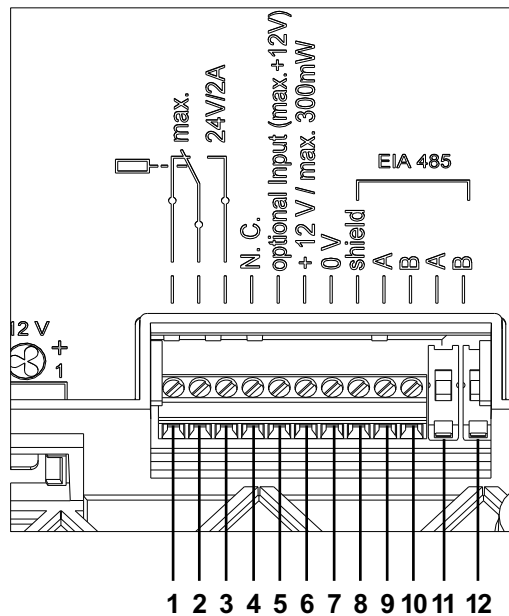


Fig. 11 Layout of terminal strip connectors

- (1) Alarm contact (break contact)
- (2) Alarm contact (center contact)
- (3) Alarm contact (NO contact)
- (4) not allocated
- (5) Input, max. 12 V (option)
- (6) Supply voltage for ext. consumer 12 V, max. 300 mW
- (7) Supply voltage for ext. consumer 0 V
- (8) Shielding EIA485 (RS485)
- (9) Lead A of EIA485 bus line and connector for termination resistor
- (10) Line B of EIA485 bus line
- (11) Line A of EIA485 bus line
- (12) Line B of EIA485 bus line

Note

Terminal strip is protected by cover. Opening the cover is possible without tools.

If the terminal screws are used, DIEHL Controls recommends using connectors on terminal strip of EIA485 bus line as follows:

- ➔ For incoming line pair:
 - ➔ terminal 9
 - ➔ terminal 10
- ➔ For outgoing line pair:
 - ➔ terminal 11
 - ➔ terminal 12

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Connection in PLATINUM network (EIA 485)

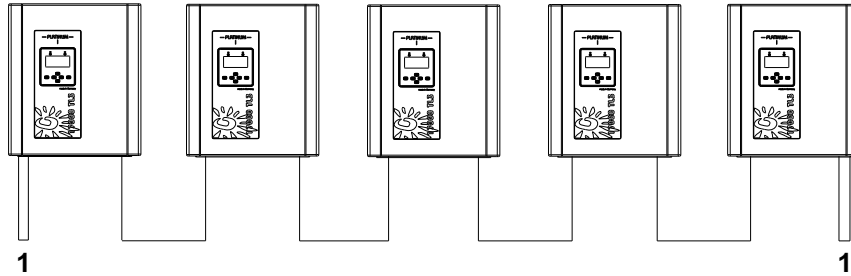


Fig. 12 Network layout

(1) Termination plug



Warning

Material damage due to interchanged lines!

- Ensure that lines A and B are allocated correctly.
- Use exclusively CAT-5 cables with twisted lines and shielding.

Establish connection to network as follows:

- Establish connection between inverters or inverters and monitoring devices on serial interface EIA485 using a cable with RJ45 plugs or cable on terminal connector (connection A and B) (see Fig. 10).
- Plug a termination plug in an inverter and/or a monitoring device on both open ends of the network.

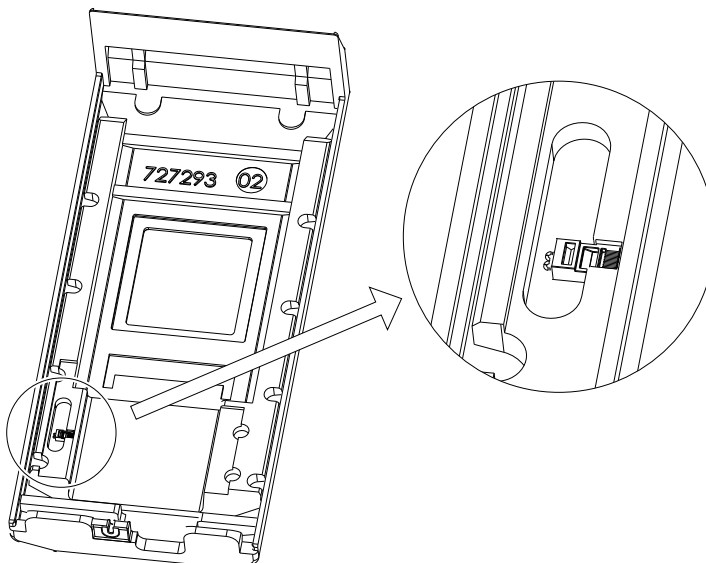


Fig. 13 Termination plug

*The termination plug is clamped into the inside of the protector and fixated with a sticker labelled **Netzwerk Terminierungsstecker / Network Termination Plug**.*

Note

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Assembly

Alarm Contact

- ➡ Connect any external indicator device to the potential-free contact.
 - When an error is detected, the contact closes and activates the indicator device (visual or acoustic warning) if necessary.
 - Setting: see menu Settings → function Alarm contact
- ➡ Use exclusively safety extra low voltage (SELV) of max. 24 V as supply voltage.

Mounting protector

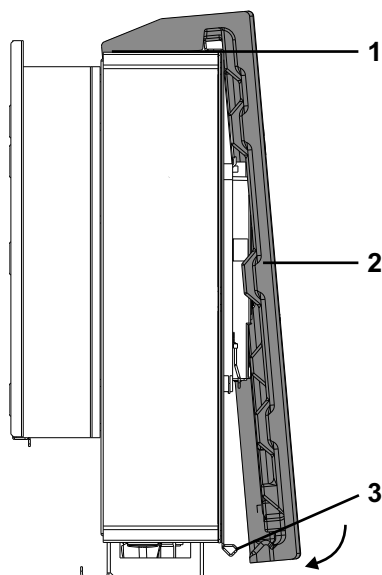


Fig. 14 Mounting protector

- (1) upper fixture
- (2) Protector
- (3) lower fixture

- ➡ Put upper part of protector 2 on upper fixture 1 and slightly push down housing.
- ➡ Slide bottom part of protector in lower fixture 3.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Start Up

6 Start Up

During the startup, several basic settings such as language selection, date and time settings are set. Information on display and operation see 7.2 Display and operation.

In order to transfer all settings correctly, it is necessary to do one complete startup.

Note

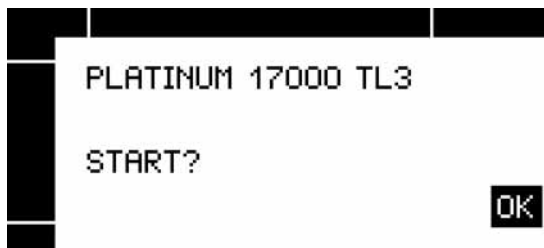
If several inverters are installed which are networked via the EIA 485 interface, the startup can be carried out at any one of the installed inverters (master programming). This inverter transmits the configuration settings to all other inverters in the network. Every inverter is automatically assigned a number at its initial startup. The free allocation of these numbers is possible in a further step.

The inverters that are not used for input entries display other screen contents depending on the menu. If no entry is possible, the inverter shows the start screen.



Start screen

If any input has been entered in the inverter, the inverter shows the start screen after a restart.



All inverters that are not operated during the configuration will show the blocking screen.

If parameters of the inverter have been adapted to special requirements of the energy provider, this is indicated on the start screen.

Note

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Start Up

After changing parameters, the inverter shows the following screen:



PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Start Up

6.1 Initial startup of inverter



Lack of function due to incomplete setting!

➔ Reconfigure country code within four hours after startup of inverter. Setting at a later time is no longer possible.

Caution

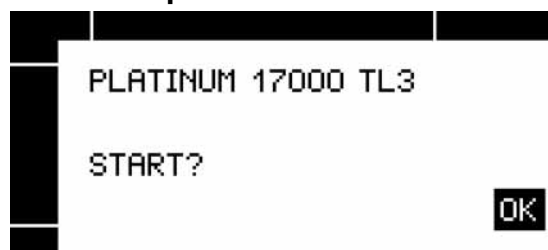
➔ Switch on the main power supply (fuse).

For inverters with DC switch:

➔ Set switch knob to ON.

The inverter is supplied with power. Configuration through the following dialogs.

Initial startup



Key	Function
OK	Start configuration process with this inverter.

Language selection



Key	Function
▲ ▼	Select language.
OK	Accept selected language.

After the configuration process has been started:

- ➔ the inverter scans the network connected via the EIA485-Bus for other inverters.
- ➔ all inverters connected to the network are blocked.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Start Up

EIA485 bus

As long as the network scan (Scanning Network) is still in process, the display shows the following screen:

```

EIA4B5
SCANNING NETWORK
NO. OF PARTICIPANTS 01
PLEASE WAIT
■■■■■■■■■■■■■■■■■■■■

```

When the network scan is completed, the display shows the number of detected bus participants. In case of a single inverter, the display shows 01.

Note

The network scan can only recognize more than 1 inverter if the inverters are correctly networked via the EIA 485 interface.

EIA485	
NO. OF PARTICIPANTS	05
NO. OF INVERTERS	04
OK	REPEAT

If the number of detected bus participants (e.g. inverters, data loggers, etc.) does not agree with the installation:

- ➡ Check connections (EIA 485 interfaces) if required.
- ➡ Select REPEAT.
- ➡ Press OK key.

If the number of detected bus participants is in agreement with the installation;

- ➡ Select OK.
- ➡ Press OK key.

In case of a single inverter:

- ➔ Screen **Country Code** appears.

In case of several networked inverters:

- ➔ Screen **Inverter numbering** appears.

Note

If one or several inverters have been replaced or not all inverters are numbered correctly, this can be corrected (see 6.2 Exchanging inverters).

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Start Up

Inverter numbering

It is possible to network several inverters via the EIA 485 interface. Inverter numbers are allocated automatically during the network scan. The inverter number is displayed in the lower left corner of the screen.

For a clearer overview, the inverters can be re-numbered for specific requirements. For instance, the inverter number can be defined according to the installation sequence.

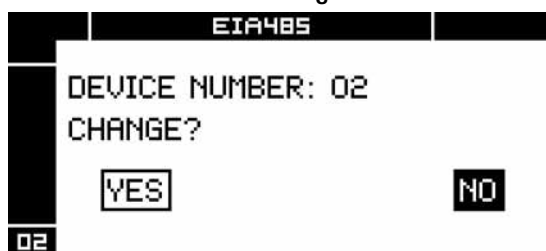
With three inverters installed, the left one can be allocated the no. 1, the middle one can be no. 2 and the right-hand one no. 3.

Example

The following screen is not displayed if only one inverter is configured.

Note

Screen Inverter numbering:



Key	Function
◀▶	YES = Allocate user-defined number to inverter. NO = Accept inverter number unchanged.
OK	Confirm selection.

If NO was selected:

- The inverter adopts the displayed number and switches to the **Country Code Setting** screen.

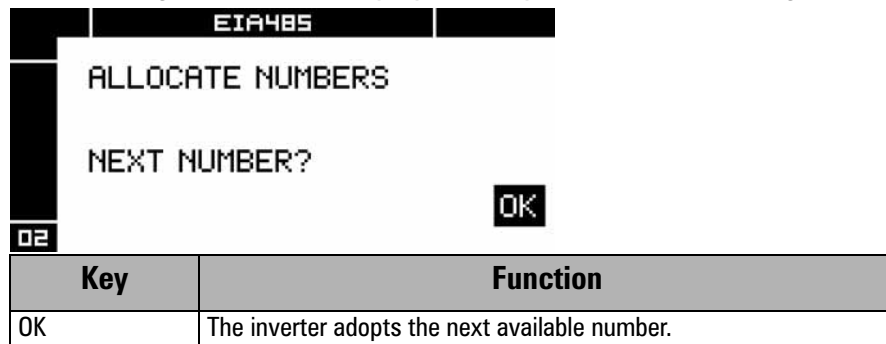
PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Start Up

If YES was selected:

→ The display shows the following screen.

Note

The following screen is not displayed if only one inverter is configured.



After OK was selected:

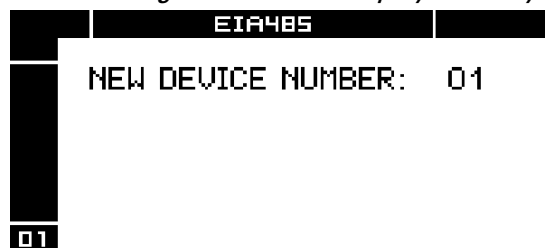
- Allocate numbers to inverters in the required sequence.
 - The inverter adopts the next available number.
 - The display shows the screen below:

Example

As soon as the OK key was pressed at the first inverter, this inverter is allocated no. 1, which is shown on the display. When the OK key was pressed at the second inverter, this inverter is allocated no. 2, etc.

Note

The following screen is not displayed if only one inverter is started up.



- Entry only at one inverter. All inverters connected to the network are blocked.
- With the number allocation completed, the inverter switches to the **Country Code** screen.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Start Up

Country code

In the menu **Country code**, the country where the inverter is operated is set. Country-specific net conditions are set automatically in dependence on the selected country.

The setting can be changed within 4 hours after the setting and the start of the feeding into the mains supply. Without feeding, the period is prolonged. After the period has expired, the setting cannot be changed anymore.

Note

```

20:05  CONFIGURATION  23.04.12
MO
< SELECT COUNTRY >
  AT - AUSTRIA
  AU - AUSTRALIA
  BE - BELGIUM
  
```

Key	Function
▲▼	Select country.
OK	Accept selection.

To select country code:

- ➞ Select desired country.
- ➞ Press OK key.

```

20:04  GERMANY  23.04.12
MO
VDE 128-1-1
VDE-AR-N 4105
BDEW-MS-RL
  
```

Key	Function
▲▼	Select standard for system.
OK	Accept selection.

System standard in Germany:

- ➞ BDEW-MS-RL for systems according to BDEW Medium Voltage Directive
- ➞ VDE-AR-N 4105 for systems according to Low Voltage Directive
- ➞ VDE 0126-1-1 for other systems

Example:

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Start Up

Inverter shows following screen:

```

┌────────── CONFIGURATION ─────────┐
│ COUNTRY SETTING                   │
│ GB                                │
│                                     │
│ ER G83/1:2008                     │
│                                     │
│                                     │
└────────── OK ─────────┘

```

➡ Press OK key.

The inverter automatically transmits the country code setting to all network participants.

If the transmission has been successful and a network is established, the inverter will show the number of configured inverters and the type of configuration.

```

┌────────── CONFIGURATION ─────────┐
│ NO. OF INVERTERS                   │
│ CONFIGURED                         │
│                                     │
│                                     │
│                                     │
└────────── OK ───────── REPEAT ───┘

```

Note

If an inverter is replaced, it is exclusively possible to change the country code of the replacement inverter.

Date

```

┌────────── DATE ─────────┐
│ DD-MM-YYYY               │
│                             │
│ ▲                         │
│ 11-08-2010               │
│ ▼                         │
│                             │
└────────── OK ─────────┘

```

Key	Function
▲▼	Increment or decrement present digit.
◀▶	Select next or previous digit.
OK	Accept date setting.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Start Up

Time



Key	Function
▲▼	Increment or decrement digit.
◀▶	Select next or previous digit.
OK	Accept time setting.

Changing date or time can result in overwriting saved data or cause gaps in data recording.

Note



To accept time setting:

- ➡ Select SELECT.
- ➡ Press OK key.

The inverter automatically shares the new time setting with all network participants.

To reset time:

- ➡ Select BACK.
- ➡ Press OK key.

After setting the time, the configuration (language, date and time) are automatically transferred to all other inverters in the network (if available).

After the data transmission, the inverter will show the menu **Time settings** (see 7.6 Settings).

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Start Up

6.2 Exchanging inverters

If one or more PLATINUM inverters in a PV system are replaced by another device, it is possible to transfer the device numbers of the replaced inverters.

During the network scan, the inverter automatically detects if devices have been replaced and provides the possibility to adjust the numbering.

It is possible to either use the old device numbers for the substitute devices or to reassign the numbers.

If inverters without a device number or device numbers that are used twice are detected during the network scan, it is possible to correct this by reassigning the numbers.

Display and operation are confined to the replaced inverters. All other inverters show the start screen.

Note

The following screen appears exclusively after exchanging an inverter.



To accept device number of a replaced inverter:

➡ Select **EXCHANGE INVERTER**.

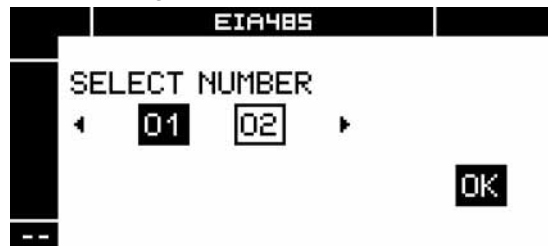
To renumber:

➡ Select **REASSIGN NUMBERS**.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Start Up

Exchanging inverters

After **EXCHANGE INVERTER** was selected, the replaced inverters show the following screen:



Key	Function
◀▶	Select device number.
OK	Accept selected device number.

On every replaced inverter:

- ➡ Select desired device number.
- ➡ Press OK key.

Inverter does another network scan.

The inverter automatically transmits the device numbers to all other inverters.



Inverter is added to PLATINUM network (EIA 485).

Inverter shows main menu.

Inverter shows device number of the replaced device in the lower left corner.



After inverter exchange:

- ➡ Set date and time.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3

Start Up

Reassigning numbers

- ➡ Reassigning numbers see 6.1 Initial startup of inverter.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Operation

7 Operation

7.1 DC switch

A banderole on the lower part of the inverter indicates where the DC disconnect is installed.

Note

The DC switch enables switching on and off the solar generator.

To engage the solar generator:

➡ Set DC disconnect to ON.

To separate solar generator from inverter:

➡ Set DC disconnect to OFF.

DIEHL Controls recommends actuating the DC switch once a year to prevent the welding of contacts.

Note

DIEHL Controls recommends switching off the grid voltage (AC) before switching off the DC switch to minimize wear and tear of the contacts.

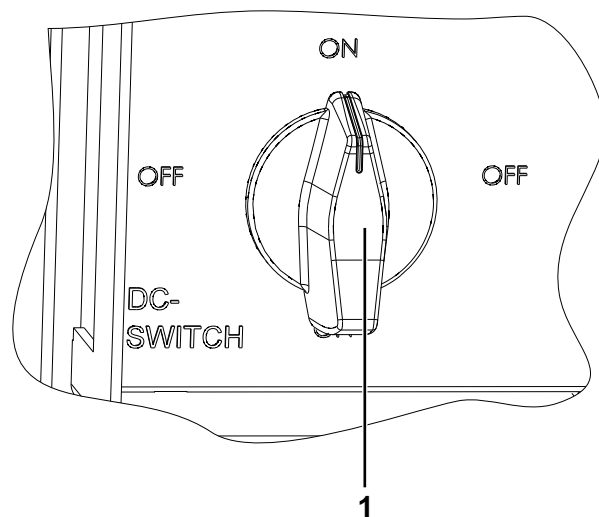


Fig. 15 DC switch

(1) DC switch

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Operation

7.2 Display and operation

Note

If there is no input for approx. 2 minutes, inverter will switch to standard display during feed.

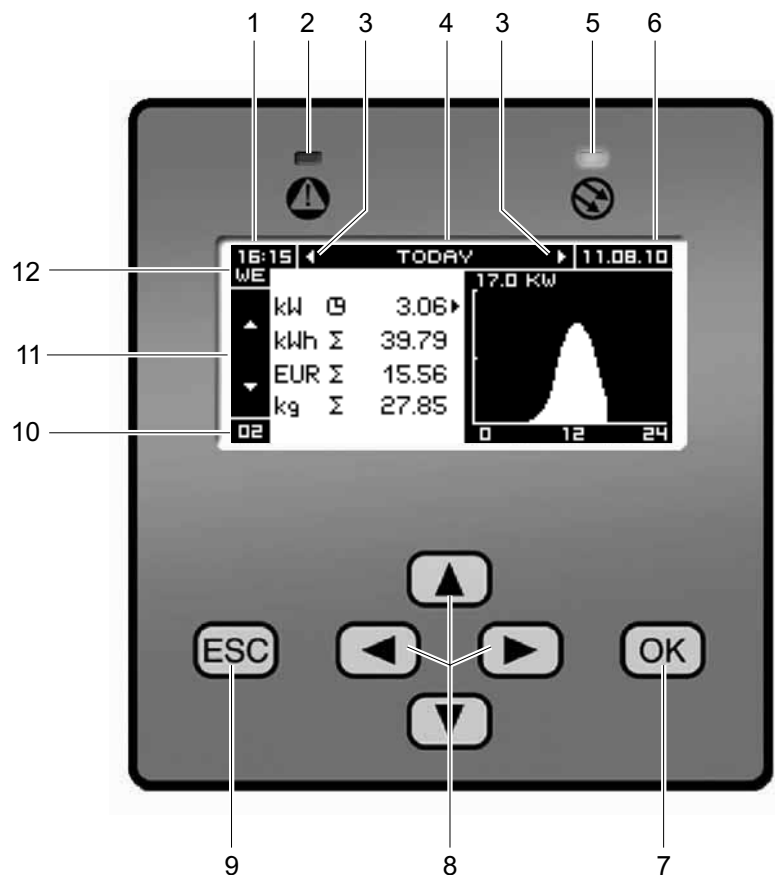


Fig. 16 Display and operation

- (1) Time
- (2) LED red
- (3) Scroll arrows horizontal
- (4) Title
- (5) LED green
- (6) Date
- (7) OK key
- (8) Navigation keys
- (9) ESC key
- (10) Device number
- (11) Scroll arrows vertical
- (12) Day

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Operation

Time

Time display in 24-hour format.

LEDs

Two LEDs indicate the operating status of the inverter.

The LED red indicates the following:

LED red

Display	Meaning
LED off	Normal operation
LED flashes	- Error - Contact for external indicator closed (depending on selected setting)

The LED green indicates the following:

LED green

Display	Meaning
LED on	Power feed on
LED flashes	Preparing for power feed
LED off	Inverter off

If both LEDs are flashing, the inverter is writing new parameter settings.

Both LEDs

Scroll arrows

The menu contains other menu items.

➡ Navigate using keys ▼ and ▲ or ◀ and ▶.

Title

Title of the selected menu.

Date

Date display in formats DD.MM.YY, MM.DD.YY or YY.MM.DD.

Adjustable in menu **Time settings**.

Keys

The functions of the keys are indicated in the tables under the illustrations.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Operation

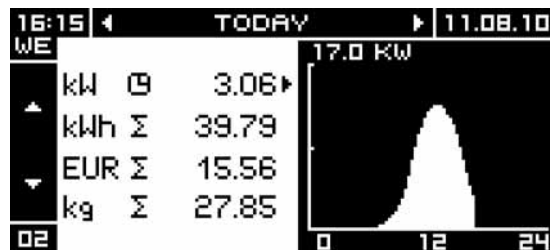
Device number

Displays the inverter number.

Day

Displays the day of the week.

7.3 Standard screen



Key	Function
◀▶	Navigate within the display period.
▲▼	Switch to screen Current .
ESC	Call up Main Menu .

Arrow to the right of the table:

→ designation of physical value displayed in graphic

Number in the top left corner of the graphic:

→ maximum value of scale

→ depending on max. DC power of the inverter

Note

The internal measuring devices are not calibrated. Small deviations of the indicated values are possible.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Operation

7.4 Main Menu



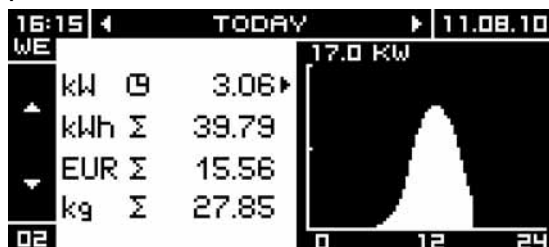
Key	Function
▲▼	Navigate in the menu.
OK	Call up the selected menu.

To return to main menu from all menus:

➡ Press ESC repeatedly.

7.5 Operation Display

The operation display shows a list of physical values and a corresponding evaluation graph. Type and value of the displayed physical value depend on the selected period. The illustration below shows an example for the period **TODAY**.



Key	Function
◀▶	Navigate within the display period.
▲▼	Switch to menu Current .
ESC	Back to Main Menu .

Arrow to the right of the table:

➡ designation of physical value displayed in graphic

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3

Operation

Units:

- W: power fed in (values < 1,000 in W, values ≥ 1,000 in kW)
- kWh or MWh: feed power for the indicated period
- EUR: earning for the indicated period (Adjustable in menu Settings.)
- kg: quantity of saved carbon dioxide (CO₂)

Number in the top left corner of the graphic:

- maximum value of scale
- dependent on inverter power

Horizontal axis in graphic:

- time scale (e. g. hours of a day)

Current

Current shows a list of the current electric values for DC and AC side.

16:09	ACTUAL	11.08.10
WE		
		DC AC L1
▲	VOLTAGE	625V 234V
▼	CURRENT	5.4A 4.7A
	POWER	3.4kW 3.3kW
02		

Key	Function
◀▶	Navigate within the display period.
▲▼	Switch to menu Today .
ESC	Back to Main Menu .

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Operation

Physical values

The following physical values are indicated:

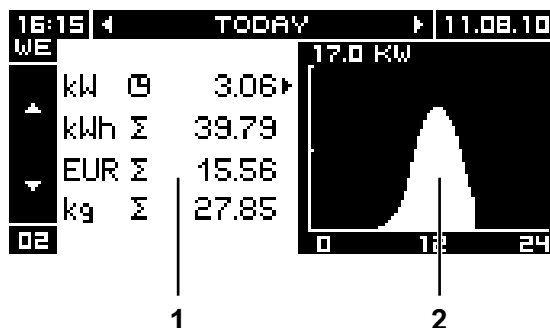
- feed power in W (graphically depicted in the periods **TODAY** and **YESTERDAY**)
- feed energy in kWh or MWh (graphically depicted as columns in the periods **WEEK**, **MONTH** and **YEAR**)
- earning in country-specific currency
 - Values > 999,000 are displayed as a factor
 - Example: 1.234.567 € is displayed as 1.234E6
- CO₂ emission reduction in kg or t
- DC and AC voltage
- DC and AC
- DC and AC power

Period

The following display periods can be selected:

- Today
- Yesterday
- Current week
- Previous week
- Current month
- Previous month
- Current year
- Previous year
- Since startup

The values displayed by the inverter may differ from the readings of calibrated electricity meters. **Note**



- (1) Physical values
- (2) Graphic of physical value

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3

Operation

Physical values:

- Designation of the physical value
- Current value (☐)
- Peak value (⬆)
- Cumulative value (Σ)

Graphic of physical value:

- Day: in hours (0 - 24)
- Week: one column per day (Mo. - Su.)
- Month: one column per day
- Year: one column per month (Jan. – Dec.)

Note

Type and value of the displayed physical value depend on the selected period.

*Graphic: For the display periods **TODAY** and **YESTERDAY**, the progress of the feed power is displayed. For all other display periods the feed energy per time interval is displayed.*

*Table: For the display period **TODAY**, the current power value is displayed. For all other periods, the maximum value is displayed.*

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Operation

7.6 Settings

In the menu **Settings** the following settings are possible:

- Time settings
- Language
- Alarm volume
- Alarm contact function
- LCD
- Earning
- System
- Energy meter



Key	Function
▲▼	Navigate in the menu.
ESC	Back to Main Menu .
OK	Call up the selected menu.

Time settings

In the menu **Time settings**, the following settings are possible:

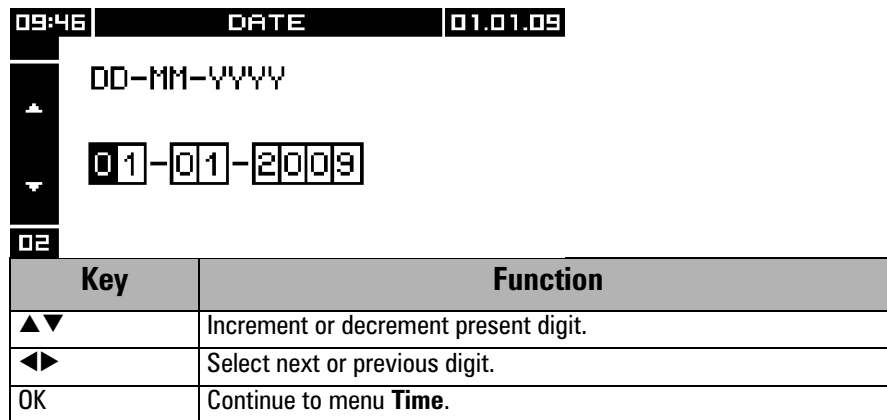
- Date
- Time
- Date format
- Daylight saving time



Key	Function
▲▼	Navigate in the menu.
◀▶	Back to menu Settings .
OK	Call up the selected menu.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Operation

Date

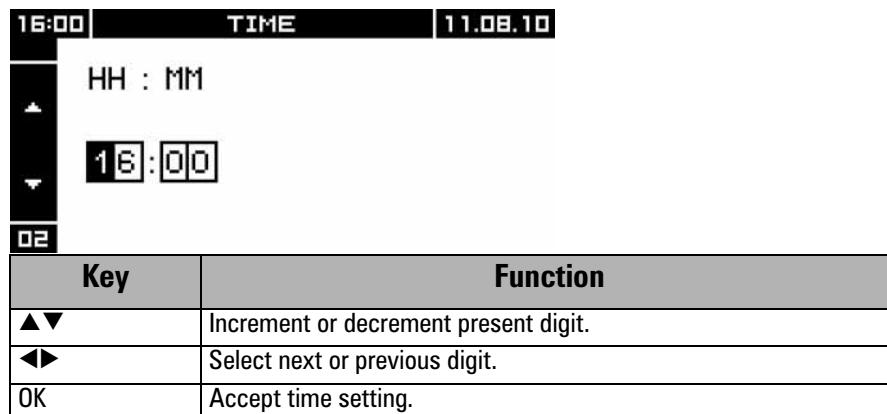


Key	Function
▲▼	Increment or decrement present digit.
◀▶	Select next or previous digit.
OK	Continue to menu Time .

Note

The date format depends on selected setting.

Time



Key	Function
▲▼	Increment or decrement present digit.
◀▶	Select next or previous digit.
OK	Accept time setting.

Note

Changing date or time can result in overwriting saved data or cause gaps in data recording.



To accept time setting:

- ➡ Select SELECT.
- ➡ Press OK key.

The inverter automatically shares the new time setting with all network participants.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Operation

To not accept time setting:

- ➔ Select CANCEL.
- ➔ Press OK key.

Date format



Key	Function
▲▼	Select format.
OK	Confirm format.
ESC	Return to menu Time setting .

Daylight saving time



Key	Function
▲▼	Navigate in the menu.
OK	Confirm selection.
ESC	Return to menu Time setting .

MANUALLY

- ➔ Manual setting of daylight saving time required.

AUTOMATIC

- ➔ Inverter sets daylight saving time automatically, depending on set country and calendar.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3

Operation

If the MANUALLY option is selected, the following screen will appear the next time the date or time is set:



To set daylight saving time:

- ➡ Select YES.
- ➡ Press OK key.

Inverter adds 1 hour to set time.

To not set daylight saving time:

- ➡ Select NO.
- ➡ Press OK key.

Inverter uses set time unchanged.

Language



Key	Function
▲▼	Navigate in the menu.
ESC	Back to menu Settings .
OK	Accept selected language.

The inverter automatically shares the new language setting with all network participants.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Operation

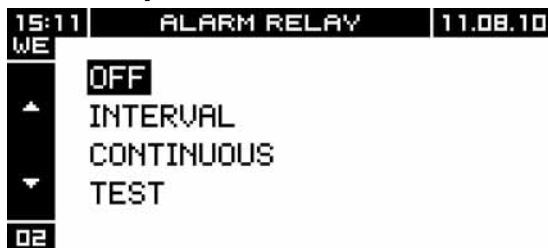
Alarm volume



Key	Function
▲▼	Increase or reduce alarm volume.
ESC	Back to menu Settings .
OK	Accept alarm volume setting.

The inverter automatically transfers the alarm volume setting to all network participants.

Alarm relay



Key	Function
▲▼	Navigate in the menu.
ESC	Back to menu Settings .
OK	Apply setting.

OFF

- ➔ The alarm relay (NO contact) is constantly open if a safety-relevant or blocking error occurs.
- ➔ Alarm contact deactivated.

INTERVAL

- ➔ The alarm relay (NO contact) opens and closes periodically if a safety-relevant or blocking error occurs.

CONTINUOUS

- ➔ If a safety-relevant or blocking error occurs, the alarm relay (NO contact) is constantly closed until the error is remedied.

TEST

- ➔ Close alarm relay temporarily while selecting a menu item.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3

Operation

LCD

16:10	LCD	11.08.10
WE		
▲	CONTRAST	30
▼	BRIGHTNESS	6
02		

Key	Function
▲▼	Increase or reduce contrast or brightness.
◀▶	Navigate between input fields.
ESC	Back to menu Settings .
OK	Apply setting.

Feed rebate

16:07	REBATE	11.08.10
WE		
▲	CURRENCY	EUR
▼	VALUE / KWH	00,391
02		

Key	Function
CURRENCY	Currency in which the rebate is displayed.
VALUE/KWH	Value for 1 kWh feed current (for rebate calculation).
▲▼	Increment or decrement present digit.
◀▶	Navigate between input fields.
ESC	Back to menu Settings .
OK	Apply setting.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Operation

System

21:04	SYSTEM	23.04.12
MO		
NAME		
PU-SYSTEM		
DESCRIPTION		
01		

Key	Function
▲▼	Increment or decrement present digit.
◀▶	Navigate between input fields.
ESC	Back to menu Settings .
OK	Apply setting.

Energy meter

The energy meter permits the metering of energy and operating hours since the start of the inverter or since the most recent reset of the energy meter.

16:47	ENERGY METER	11.08.10
WE		
SINCE	11.08.2010	
ENERGY	13.12 kWh	
TIME	3h	
BACK	RESET	
02		

Key	Function
BACK	Back to menu Settings .
RESET	Set energy meter to 0.
◀▶	Select BACK or RESET.
OK	Confirm selection.
ESC	Back to menu Settings .

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3

Operation

7.7 Information

The menu **Information** shows the following information:

- Operating data
- System data
- Inverter type
- Inverter version
- Event information

Note

*The menu **Information** merely displays the value. It is not possible to change the values.*

Operating data

16:44	ENERGY METER	11.08.10
WE		
	GENERAL	METER 2
▲	09.02.2005	11.08.2010
▼	45.862 MWh	13.12 kWh
	20500 h	3 h
02		

Key	Function
▲	Call up screen Inverter version .
▼	Call up screen System data .
ESC	Back to the Main Menu .
OK	Back to the Main Menu .

GENERAL

- Shows feed data of the inverter since the startup.
- Resetting is not possible.

METER 2

- Shows feed data of the inverter since the latest reset of meter 2.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Operation

System data

21:01	SYSTEM	23.04.12
MO		
▲	NAME	PV-SYSTEM
▼	NO. OF PARTICIPANTS	01
01	NO. OF INVERTERS	01

Key	Function
▲	Call up screen Operating data .
▼	Call up screen Inverter type .
ESC	Back to the Main Menu .
OK	Back to the Main Menu .

NAME

→ Shows the name of the PV system.

NO. OF PARTICIPANTS

→ Indicates number of network participants (e. g. inverter and monitoring devices such as PLATINUM PV-Monitor, PLATINUM ViewMaster and PLATINUM Webmaster).

NO. OF INVERTERS

→ Shows the number of inverters in the network.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Operation

Inverter type

20:59	INFORMATION	23.04.12
MO		
▲	TYPE	13000 TL3
		ENS3-DE
▼	S/N	1007.120514001
		VDE 0126-1-1:2006-02
01		+A1:2011-06

Key	Function
▲	Call up screen System data .
▼	Call up screen Inverter version .
ESC	Back to the Main Menu .
OK	Back to the Main Menu .

TYPE

➔ Indicates inverter type.

S/N

➔ Indicates inverter serial number.

Note

When contacting the PLATINUM service, have the serial number ready.

*After changing the inverter network parameters, the display shows **USER-DEFINED NETWORK MONITORING**.*

16:18	INFORMATION	11.08.10
WE		
▲	TYPE	PLATINUM 17000 TL
		5P-DE
▼	S/N	8007.100405002
		USER DEFINED
02		GRID MONITORING

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Operation

Inverter version

15:40	INFORMATION	11.08.10
WE		
▲	FIRMWARE	06 05 04
▼	HARDWARE	--
	EIA485	V 4.11
	BL	02 -- --
02		

Key	Function
▲	Call up screen Inverter type .
▼	Call up screen Operating data .
ESC	Back to the Main Menu .
OK	Back to the Main Menu .

FIRMWARE

→ Shows the inverter firmware version.

HARDWARE

→ Shows the inverter hardware version.

EIA485

→ Shows the data bus software version.

BL

→ Shows version of bootloader software.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Operation

7.8 Error display

The error displays are meant for the user. Any detected errors are displayed with a slight delay. In case of an error, the inverter displays the error type and error code. Error causes and measures see 12 Troubleshooting.

Three types of errors:

- ➔ fatal errors
- ➔ blocking errors
- ➔ non-blocking errors

Fatal errors

 Warning	Destruction of the inverter due to fatal errors! ➔ Switch off AC. ➔ Switch off DC. If available: ➔ Set DC switch to 0. ➔ Notify PLATINUM service.
--	---

16:46	ERROR	11.08.10
WE		
	ERROR	F0930-13
	DISCONNECT DC AND AC	
	CHECK CONNECTIONS!	
02		

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3

Operation

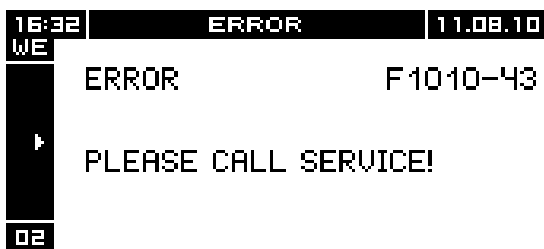
Blocking errors



Danger

Danger to life due to electric shock!

➔ Have the inverter opened exclusively by the PLATINUM service or service partners authorized by DIEHL Controls.



In case of a blocking error:

- ➔ the inverter is permanently off.
 - ➔ the inverter gives a visual alarm (red LED flashing).
 - ➔ the inverter gives an acoustic alarm.
 - ➔ the inverter closes the alarm contact.
- Setting: see screen **Settings** → **function Alarm contact**.

Exclusively service personnel can clear the blocking error and switch the inverter back on.

To stop the acoustic alarm:

- ➔ Press any key.

To delete the error display:

- ➔ Press ESC key.

If the acoustic alarm is activated:

- ➔ Inverter switches off acoustic alarm.

- ➔ Press key ESC again.

Inverter shows start screen.

Red LED will flash as long as the error persists.

Operation

Non-blocking errors and warnings

16:29	ERROR	11.08.10
WE		
	ERROR	F2100-73
02		

Key	Function
OK	Clears the error display
ESC	Leave error display.

Non-blocking errors are of a transient nature (e.g. mains voltage surge). The inverter remains switched off until the error cause no longer persists.

When the error has been remedied, the inverter automatically switches to normal operation.

Until the error is resolved:

- ➔ The inverter displays the screen **Non-blocking error**.
- ➔ The red LED is flashing.
- ➔ Delete error display with key ESC.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3

Service

8 Service

8.1 Service Menu

The menu Service offers the following information and functions:

- display event list
- display parameters
- reconfiguration
- display startup date
- display feed meter
- display and set grid support
- set power reduction

To display other parameters or to change the parameters is exclusively possible with a separate service tool.

Note

To call up the service menu:

- ➔ Select item **Time settings** in main menu and press keys ◀ and ▶ simultaneously for approx. 3 s.

20:28	SERVICE MENU	23.04.12
MO		
	EVENT LIST	
▲	PARAMETER	
▼	INITIAL OP. DATE	
	METER	
01		

Key	Function
▲▼	Navigate in the menu.
ESC	Return to standard display.
OK	Call up the selected menu.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Service

Event list

20:25	EVENT LIST	23.04.12
MO		
▲	001 23.04.12 20:24 E6750-D1	
	002 23.04.12 20:24 E4010-D1	
	003 23.04.12 20:24 E6750-D1	
▼	004 23.04.12 20:24 E4010-D1	
	005 23.04.12 20:24 E6750-D1	
01		

Key	Function
▲▼	Navigate within the event list.
ESC	Back to the Service menu .

Note

The inverter shows the last 100 detected events.

Explanations about the events see 12.1 Table of events.

Have the event code and the serial number ready when contacting the PLATINUM service.

Parameters

In some supply areas, the values for supply voltage and frequency may differ temporarily or permanently from the factory settings. It is possible to adapt the PLATINUM inverter to these values. Contact the PLATINUM service for more information.

The screens **Parameter** show the set parameters, the country code and the applicable directive.

Note

The parameters can exclusively be modified by certified persons with the PLATINUM service tool.

20:50	PARAMETER	23.04.12
MO		
▲	0000 Parameter	
◀▶	0000.0 Product	
▼	0000.1 Structure	
	0000.2 Content	

Key	Function
▲▼	Navigate within the parameter menu.
OK	Open menu of parameter group selected.
ESC	Back to the Service menu .

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Service

14:28	PARAMETER	10.05.11
DI		
	1007 VoltMon	
▲	1007.0 Limit	
◀▶	1007.1 TripTime	
▼		
02		

Key	Function
▲▼	Navigate within menu of parameter group.
◀▶	Change superordinate parameter group.
OK	Select menu item.
ESC	Switch to superordinate menu.

14:29	PARAMETER	10.05.11
DI		
	1007 VoltMon	
	1007.0 Limit	
◀▶	1007.0.0 U1_Max	
	264.5 V	
02		

Key	Function
▲▼	Switch between individual values.
◀▶	Navigate within individual values.
ESC	Switch to superordinate menu.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Service

Reconfiguration

Inverter shows Start screen and carries out startup process again (see 6.1 Initial startup of inverter).

Note

A reconfiguration of the country is only possible within the first four hours after the connection. Afterwards, this menu item is blocked.

Start Up

16:02	INITIAL OP. DATE	11.08.10
WE	DD-MM-YYYY	
	09.02.2010	
02		

Key	Function
OK	Back to the Service menu .
ESC	Back to the Service menu .

Shows startup date.

Note

The menu is only a display. Values cannot be changed.

*The date format depends on the selected setting in menu **Time settings**.*

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Service

Meter

20:53	ENERGY METER	23.04.12
MO		
	SINCE	01.01.2009
↔	ENERGY	0.10 kWh
	TIME	0 h
	BACK	RESET
01		

Key	Function
OK	Back to the Service menu .
ESC	Back to the Service menu .

The menu is only a display. Values cannot be changed.

Inverter numbering

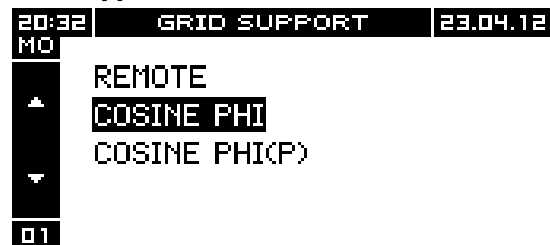
To change the inverter numbering after startup:

- ➡ Select **Reconfiguration** in the Service Menu.
- ➡ Change inverter numbers see 6.1 Initial startup of inverter, subsection allocating inverter numbers.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3

Service

Grid support



Key	Function
▲▼	Navigate in the menu.
ESC	Back to the Service menu .
OK	Confirm setting.

REMOTE

- ➔ External device (PLATINUM WebMaster) sets cosinus phi or reactive power.

COSINUS PHI

- ➔ Set cosinus phi according to the energy provider's specifications.

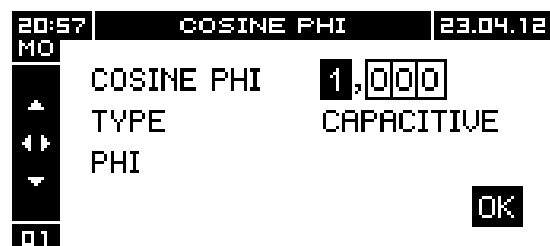
COSINUS PHI(P)

- ➔ Set cosinus phi according to specifications backed up in the PLATINUM inverter.

Note

After pressing the OK key, the inverter loads the backed-up data.

Setting cosinus phi



Key	Function
▲▼	Increment or decrement present digit. Switch between CAPACITIVE and INDUCTIVE.
◀▶	Select next or previous digit. In the beginning or end of a value: Select next or previous line.
ESC	Back to the Service menu .
OK	Confirm settings.

Range of cosinus phi values:

- ➔ 0,707 – 1

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Service

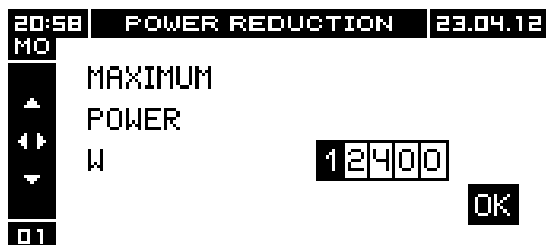
Power reduction

Legal requirements in some countries demand power reduction of the AC power fed in.

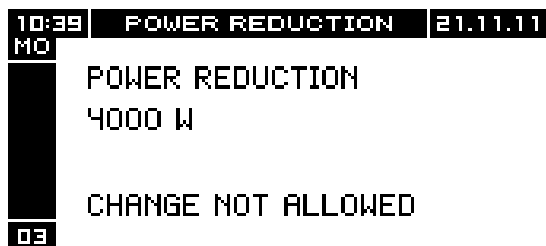
- ➞ If necessary, reduce inverter power.

Example:

- ➞ Reduce system with 17,715 Wp permanently by 70%.
- ➞ Setting $17,715 \text{ Wp} * 0.7 = 12,400 \text{ W}$



Key	Function
▲▼	Increment or decrement present digit.
◀▶	Select next or previous digit.
ESC	Back to the Service menu .
OK	Confirm settings.



To accept power reduction setting:

- ➞ Select SELECT.
- ➞ Press OK key.

To not accept power reduction setting:

- ➞ Select CANCEL.
- ➞ Press OK key.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3

Maintenance

9 Maintenance



Danger

Danger to life due to electric shock!

- ➔ Have the inverter opened exclusively by the PLATINUM service or service partners authorized by DIEHL Controls.



Danger

Danger to life do to DC and AC high voltage!

- ➔ Wear insulating protective clothing and face protection.
- ➔ Have any cleaning or maintenance works carried out exclusively by trained personnel.
The qualified specialist personnel require a license from the relevant energy providers.

Prior to every maintenance work or cleaning:

- ➔ Switch off the mains power supply (fuse).
- ➔ Set switch knob of DC disconnecter to OFF.
- ➔ Condensers require up to 30 min to discharge. Do not touch connectors (DC/AC) for at least 5 minutes after switching off.
- ➔ Ensure that the DC cables are de-energized.

In case of a PLATINUM inverter without DC switch:

- ➔ Pull the plugs in the following order:
 1. AC side
 2. DC side

9.1 Maintenance

The inverter is maintenance-free.

Note

DIEHL Controls recommends having checked the inverters interior for dust and humidity by qualified specialists every 3 – 4 years.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Maintenance

9.2 Cleaning

To assure the cooling, regularly:

- Clean ventilation openings with:
 - ➔ vacuum cleaner
 - ➔ soft brush
 - ➔ compressed air

If available:

- Remove plants from inverter.
- Remove any animals nesting on or in inverter or inside cooling element.

In a dusty environment:

- Check fan and cooling elements regularly and clean them if necessary.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3

Placing out of operation

10 Placing out of operation

10.1 Dismounting



Danger

Danger to life due to DC and AC high voltage!

- Wear insulating protective clothing and face protection.
 - Have inverters uninstalled by qualified specialists only. The qualified specialist personnel require a license from the relevant energy providers.
 - Switch off the mains power supply (fuse).
 - Set switch knob of DC disconnecter to OFF.
 - Do not touch terminals (DC/AC) for at least 5 minutes after disconnection (discharging time of capacitors).
 - Ensure that the DC cables are de-energized.
- In case of a PLATINUM inverter without DC switch:
- Pull the plugs in the following order:
 1. AC side
 2. DC side



Danger

Danger to life due to inverter dropping from the wall!

- Use mounting elements appropriate for the mounting wall and the weight of the inverter unit.
- Wear protective footwear when mounting and dismantling inverters.

To disconnect the inverter:

- Switch off the mains power supply (fuse).
- Remove cable from AC port.
- Set DC disconnecter to OFF and pull plug connector of input voltage (DC side).
- Disconnect remaining connectors as required.

Dismount the inverter as follows:

- Loosen and remove mounting screws.
- Pull inverter forward from mounting fixture and down out of upper hook row.

Remove the mounting fixture as follows:

- Unscrew the mounting fixture.
- Insert the mounting fixture at the back of the inverter.
- Secure wall mounting with locking screws.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3

Placing out of operation

In case of a return consignment:

- ➡ Pack the inverter in the packaging of the replacement.

If a single inverter is returned:

- ➡ Demand additional packaging from Diehl-Controls or reuse it.

Packaging

Pack inverter as follows:

- ➡ Pack inverter in original box.
- ➡ Pack inverter into box upside down.
- ➡ Seal box sufficiently for transport.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3

Disposal

11 Disposal



- ➔ Dispose of packaging and consumed parts according to the rules and regulations applicable in the country where the device was installed.
- ➔ Do not dispose of the PLATINUM inverter in the household waste.

Note

DIEHL Controls takes back PLATINUM inverters completely.

It is possible to dispose of PLATINUM inverter through communal disposal of electrical appliances.

12 Troubleshooting

12.1 Table of events

To isolate the error, have the following information ready when calling Service:

- ➔ displayed error number (No.)
- ➔ serial number of the inverter (see 7.7 Information)

Note

A complete event list is available in the download area of our website www.diehl.com/photovoltaics.

No.	Meaning	Measure
100	general system error	Separate inverter from DC generator and mains. Wait until inverter is de-energized (approx. 3 min). Re-engage inverter. If event occurs repeatedly: Contact PLATINUM service. Have error code and serial number ready.
1000 1250 1310	internal system error	Separate inverter from DC generator and mains. Wait until inverter is de-energized (approx. 3 min). Re-engage inverter. If event occurs repeatedly: Contact PLATINUM service. Have error code and serial number ready.
1340	Country code not yet set	Set country code. If necessary, reconfigure in service menu.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3

Troubleshooting

No.	Meaning	Measure
1350 1370 1381 1382 1500	internal system error	Separate inverter from DC generator and mains. Wait until inverter is de-energized (approx. 3 min). Re-engage inverter. If event occurs repeatedly: Contact PLATINUM service. Have error code and serial number ready.
1800	Feed lock active	
2000	Inverter is temporarily separated from mains and will automatically return to normal operation once error has been cleared.	Confirm error. If event occurs repeatedly: Contact PLATINUM service. Have error code and serial number ready.
2500 2510 to 2512	internal communication error	Separate inverter from DC generator and mains. Wait until inverter is de-energized (approx. 3 min). Re-engage inverter. If event occurs repeatedly: Contact PLATINUM service. Have error code and serial number ready.
2600	Connection PLATINUM network (EIA 485) faulty	Check connections PLATINUM network (EIA 485).
2610	internal memory error	Separate inverter from DC generator and mains. Wait until inverter is de-energized (approx. 3 min). Re-engage inverter. If event occurs repeatedly: Contact PLATINUM service. Have error code and serial number ready.
2611 2620	internal memory error	Separate inverter from DC generator and mains. Wait until inverter is de-energized (approx. 3 min). Re-engage inverter. If event occurs repeatedly: Contact PLATINUM service. Have error code and serial number ready.
2980	self-test error (exclusively with certain country codes, e.g. Italy)	Self-test exclusively possible during feed.
4010	Information reset. Only added to event list.	only for information no action necessary
4100	Time has been changed manually (on display).	only for information no action necessary
4110	Time has been changed via PLATINUM network.	only for information no action necessary
4120 4130	Parameters have been changed.	only for information no action necessary

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3

Troubleshooting

No.	Meaning	Measure
1350 1370 1381 1382 1500	internal system error	Separate inverter from DC generator and mains. Wait until inverter is de-energized (approx. 3 min). Re-engage inverter. If event occurs repeatedly: Contact PLATINUM service. Have error code and serial number ready.
1800	Feed lock active	
2000	Inverter is temporarily separated from mains and will automatically return to normal operation once error has been cleared.	Confirm error. If event occurs repeatedly: Contact PLATINUM service. Have error code and serial number ready.
2500 2510 to 2512	internal communication error	Separate inverter from DC generator and mains. Wait until inverter is de-energized (approx. 3 min). Re-engage inverter. If event occurs repeatedly: Contact PLATINUM service. Have error code and serial number ready.
2600	Connection PLATINUM network (EIA 485) faulty	Check connections PLATINUM network (EIA 485).
2610	internal memory error	Separate inverter from DC generator and mains. Wait until inverter is de-energized (approx. 3 min). Re-engage inverter. If event occurs repeatedly: Contact PLATINUM service. Have error code and serial number ready.
2611 2620	internal memory error	Separate inverter from DC generator and mains. Wait until inverter is de-energized (approx. 3 min). Re-engage inverter. If event occurs repeatedly: Contact PLATINUM service. Have error code and serial number ready.
2980	self-test error (exclusively with certain country codes, e.g. Italy)	Self-test exclusively possible during feed.
4010	Information reset. Only added to event list.	only for information no action necessary
4100	Time has been changed manually (on display).	only for information no action necessary
4110	Time has been changed via PLATINUM network.	only for information no action necessary
4120 4130	Parameters have been changed.	only for information no action necessary

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3

Troubleshooting

No.	Meaning	Measure
4140	Country codes have been set.	only for information no action necessary
4150 4160	Power reduction has been activated externally.	only for information no action necessary
4230	self-test successful (exclusively with certain country codes, e.g. Italy)	only for information no action necessary
4410 4420	Software update completed.	only for information no action necessary
4430	Configuration has been changed.	only for information no action necessary
4500	information reset (e. g. after power failure) Only added to event list.	only for information no action necessary
4600 4610	Software update completed.	only for information no action necessary
4990	Information	only for information no action necessary
6000 to 6003	internal system error	Separate inverter from DC generator and mains. Wait until inverter is de-energized (approx. 3 min). Re-engage inverter. If event occurs repeatedly: Contact PLATINUM service. Have error code and serial number ready.
6004	Country code not yet set	Set country code. If necessary, reconfigure in service menu.
6005 to 6008	internal system error	Separate inverter from DC generator and mains. Wait until inverter is de-energized (approx. 3 min). Re-engage inverter. If event occurs repeatedly: Contact PLATINUM service. Have error code and serial number ready.
6009	internal system error	Performing software update.
6010 to 6014	internal system error	If occurring during startup: Let fault persist for 2 – 3 hours. If event occurs repeatedly: Contact PLATINUM service. Have error code and serial number ready.
6015	Insulation resistance too low	Check insulation of generator to earth (e.g. damage to cable caused by animals, crushed cables, damaged modules).

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3

Troubleshooting

No.	Meaning	Measure
6016	unknown system error	Separate inverter from DC generator and mains. Wait until inverter is de-energized (approx. 3 min). Re-engage inverter. If event occurs repeatedly: Contact PLATINUM service. Have error code and serial number ready.
6017 to 6022	internal system error	Separate inverter from DC generator and mains. Wait until inverter is de-energized (approx. 3 min). Re-engage inverter. If event occurs repeatedly: Contact PLATINUM service. Have error code and serial number ready.
6023 6024	internal system error	Check DC voltage.
6025	internal system error	Separate inverter from DC generator and mains. Wait until inverter is de-energized (approx. 3 min). Re-engage inverter. If event occurs repeatedly: Contact PLATINUM service. Have error code and serial number ready.
6026 to 6039	internal system error	Separate inverter from DC generator and mains. Wait until inverter is de-energized (approx. 3 min). Re-engage inverter. If event occurs repeatedly: Contact PLATINUM service. Have error code and serial number ready.
6040	voltage above limit	Check grid voltage. If grid voltage is within normal range: Contact PLATINUM service. Have error code and serial number ready.
6041	voltage below limit	Check grid voltage. If grid voltage is within normal range: Contact PLATINUM service. Have error code and serial number ready.
6042	voltage on phase conductor above limit	Check grid voltage. If grid voltage is within normal range: Contact PLATINUM service. Have error code and serial number ready.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3

Troubleshooting

No.	Meaning	Measure
6043	voltage on phase conductor below limit	Check grid voltage. If grid voltage is within normal range: Contact PLATINUM service. Have error code and serial number ready.
6044	Grid error	Check grid frequency and grid voltage. If grid frequency and grid voltage are within normal range: Contact PLATINUM service. Have error code and serial number ready.
6045	Grid frequency error	Check grid frequency and grid voltage. If grid frequency and grid voltage are within normal range: Contact PLATINUM service. Have error code and serial number ready.
6046	Insulation error	Check insulation of generator to earth (e.g. damage to cable caused by animals, crushed cables, damaged modules).
6047 to 6050	temperature too high	Check ventilation (cooling fins dirty/blocked, air circulation blocked). Choose cooler location for installation (shade, ventilation etc.). Inverter will automatically resume operation after cooling down.
6051 6052	Grid error	Check grid frequency and grid voltage. Contact service if grid frequency is within normal range.
6053	Grid error	Check grid frequency and grid voltage. If grid frequency and grid voltage are within normal range: Contact PLATINUM service. Have error code and serial number ready.
6054 6055	internal system error	Separate inverter from DC generator and mains. Wait until inverter is de-energized (approx. 3 min). Re-engage inverter. If event occurs repeatedly: Contact PLATINUM service. Have error code and serial number ready.
6056 6057	internal system error	Check DC voltage.
6058 to 6063	internal system error	Separate inverter from DC generator and mains. Wait until inverter is de-energized (approx. 3 min). Re-engage inverter. If event occurs repeatedly: Contact PLATINUM service. Have error code and serial number ready.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3

Troubleshooting

No.	Meaning	Measure
6064	Grid frequency error	Check grid frequency and grid voltage. If grid frequency and grid voltage are within normal range: Contact PLATINUM service. Have error code and serial number ready.
6065	incorrect country code	Set country code. If necessary, reconfigure in service menu.
6066 6067	Parameter settings do not match grid voltage.	Set country code. If necessary, reconfigure in service menu.
6068	Parameter settings do not match grid frequency.	Set country code. If necessary, reconfigure in service menu.
6069	internal system error	Separate inverter from DC generator and mains. Wait until inverter is de-energized (approx. 3 min). Re-engage inverter. If event occurs repeatedly: Contact PLATINUM service. Have error code and serial number ready.
6070	Time of day not set.	Set time.
6071	Time of day not set.	Set time.
6072	wrong firmware version	only for information Update LT bootloader as soon as possible.
6073 to 6076	Information	only for information no action necessary
6999	unknown error code	Contact PLATINUM service. Have error code and serial number ready.

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3

Technical Data

13 Technical Data

Input characteristics	11000 TL3	13000 TL3	17000 TL3	20000 TL3
Maximum PV power	11,000 Wp	13,600 Wp	18,100 Wp	21,200 Wp
Maximum DC power	10,300 W	12,800 W	16,900 W	19,650 W
Maximum DC voltage	1,000 V			
PV voltage range, M P P T	380 – 850 V	420 – 850 V	445 – 850 V	480 – 850 V
Maximum input current	29.0 A	30.0 A	38.5 A	41.0 A
Number of string inputs	4		6	
Number of MPP trackers	1			
DC section switch device	DC disconnect, integrated in the appliance			
Reverse battery protection	yes			
Earth leakage protection	Insulation check			
Output characteristics				
Maximum AC power	10,000 W	12,400 W	16,500 W	19,200 W
Nominal AC power rating	10,000 W	12,400 W	16,500 W	19,200 W
Nominal AC intensity	14.5 A	18.0 A	23.9 A	27.8 A
Maximum AC intensity	18.0 A	18.0 A	29.0 A	29.0 A
Feed operation starts at	20 W			
Grid output voltage range	3AC 400 V + N (+/-20%)			
Mains frequency	50 Hz / 60 Hz (+/- 5%)			
Short-circuit proof	yes			
cos phi	0.9i – 0.9c			
Earth leakage protection	AFI			
Internal consumption at night	below 2.5 W			
Interfaces				
DC input	Multicontact MC4			
AC output	Phoenix plug connector (included)			
PLATINUM network	EIA 485, 2x RJ45 Western Modular add. plug connector with terminal screws			
Service interface	EIA 232, SubD 9-pole socket, USB			
Potential-free contact	1 change-over contact, maximum 24 V _{AC} /2A, plug connector with terminal screws			

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3

Technical Data

Appliance data	11000 TL3	13000 TL3	17000 TL3	20000 TL3
Maximum degree of efficiency	98.0%		98.2%	
European efficiency	97.4%	97.5%	97.8%	97.8%
Working temperature range		-25 °C to +55 °C		
Maximum temperature during nominal power	+50 °C		+40 °C	
Storage temperature		-20 °C to +70 °C		
Degree of protection (except digital interface)		IP 65 according to DIN EN 60529		
Dimensions		H 626 mm x W 543 mm x D 281 mm		
Weight		39 kg		40 kg
Circuit concept	Transformerless 3-phase high-performance topology, ENS according to VDE 0126-1-1			
Optical display		Full graphic LCD 170 x 76 pixels		
Integrated datalogger		Storage capacity sufficient for 30 yrs operating time		

Note *Technical data apply to maximum height above MSL of 2,000 m.*

PLATINUM Inverter 11000 TL3, 13000 TL3, 17000 TL3, 20000 TL3 Manufacturers' Warranty of Diehl AKO Stiftung & Co. KG

14 Manufacturers' Warranty of Diehl AKO Stiftung & Co. KG

Limited manufacturer's guarantee of Diehl AKO Stiftung & Co. KG for PLATINUM string inverter TL, S, H, for the third to the tenth year after installation/start of operation valid for PLATINUM string inverters from the delivery date February 01, 2011.

(Address: D-88239 Wangen im Allgäu, Pfannerstraße 75, Federal Republic of Germany)

1. Products entitled to guarantee

The present manufacturer's guarantee applies to the mains-coupled string inverters for photovoltaic systems of the PLATINUM series manufactured by Diehl AKO Stiftung & Co. KG ("Diehl AKO"), to the extent that they can be proven to have been acquired from Diehl AKO or an authorised wholesale trader, specialist trader or specialist installation company as a new appliance and delivered from February 01, 2011 ("Warranted Products"). Proof shall be deemed rendered if the original of an invoice is presented to Diehl AKO documenting delivery of a Warranted Product to a Warranted User and additionally the original manufacturer's declaration of Diehl AKO has been attached to the name plate of the Warranted Product.

2. Beneficiaries from this manufacturer's guarantee

Diehl AKO grants this manufacturer's guarantee only to users who can be proven to have acquired a Warranted Product and operate it themselves ("operator entitled to guarantee"). Traders of any kind and trading levels shall not acquire any kind of rights and claims from this manufacturer's guarantee against Diehl AKO.

3. Origination of the guarantee

The manufacturer's warranty is to be understood as an offer from Diehl AKO directly to the Warranted User for conclusion of a guarantee contract under the terms and conditions of the entire manufacturer's guarantee. The guarantee contract shall automatically originate directly between Diehl AKO and the Warranted User with acquisition of a Warranted Product, insofar as the Warranted User does not challenge the origination of the guarantee contract towards Diehl AKO in writing within two (2) weeks of acquisition of a Warranted Product.

4. Relationship of the manufacturer's guarantee to other claims

This manufacturer's guarantee grants the Warranted User within its scope and according to its terms and conditions claims against the manufacturer of the Warranted Product which are in addition to the warranty claims of the Warranted User against the respective vendor of the inverter. Warranty rights against the vendor in question and claims

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from a statutory product liability remain unaffected by this manufacturer's warranty.

5. Duration and claiming of the manufacturer's guarantee

The manufacturer's guarantee shall apply to Guarantee Cases (in the meaning of Section 6 hereinafter) of the Warranted Products which can be proven to occur between the start of the twenty-fifth and the end of the one hundred and twentieth month after installation and start of operation of a Warranty Product by the Warranted User ("Guarantee Period"). This Guarantee Period shall end no later than 132 months after the production date of the Warranted Product stated on the name plate. For Warranted Products repaired or replaced by Diehl AKO, the manufacturer's guarantee shall apply until the expiry of the original Guaranty Period. Statutory and/or contractual warranty claims of any kind which can occur during a statutory or contractual warranty period can not be derived from the present manufacturer's guarantee.

All and any claims from the manufacturer's guarantee shall be made in writing to Diehl AKO by the Warranted User to Diehl AKO within the Guaranty Period. Such guarantee claims can be submitted via an authorised specialist trader, wholesale trader or specialist installation company.

6. Defects of the products entitled to guarantee covered by manufacturer's guarantee (Guarantee Cases)

This manufacturer's guarantee shall apply to defects of Warranted Products occurring during the Guarantee Period for which Diehl AKO is responsible and which do not inconsiderably impair or reduce their feed function (in the sense of the output performance of the Warranted Products registered on the AC side - by the feed current meter - "Feed Performance") ("Guarantee Cases"). Defects in the aforementioned sense shall exclusively be defects in quality of the products which entitle to warranty in the meaning of § 434 sub-section 1 German Civil Code, resulting in a not inconsiderable reduction of the effective Feed Performance compared with the performance data contained in the installation and operating instructions for the string inverters for photovoltaic systems of the PLATINUM series affected. In addition, a Guarantee Case is excluded the extent as an exclusion situation in the meaning of the following Section 8 exists, occurs or has occurred.

7. Rights under the manufacturer's warranty - Damage and costs not covered

If a Guaranty Case occurs there shall be a free-of-charge repair of the Warranted Product or a free-of-charge replacement by a product with a maximum output performance on the AC side coming closed to the output of the Warranted Product("Guarantee Claims").

Repairs or replacement shall be exclusively be done in the Diehl AKO factory in Wangen im Allgäu. Transport to Diehl AKO must be in the

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original packaging or at least equivalent packaging. If repair or replacement is done at a deviating location determined by the Warranted User's request, Diehl AKO can comply with this; in such a case, the Warranted User shall bear the travelling expenses and the additionally necessary working time according to the standard rates of Diehl AKO.

Any claims from this manufacturer's guarantee exceeding repair or replacement free of charge are excluded, in particular claims to defect-induced economic damage such as loss of profits including remuneration for (even temporarily) failed or reduced mains feeding, installation and dismantling costs and costs of fault diagnostics.

If no Guaranty Case is established on the Warranted Product which has been delivered or if no claim from this manufacturer's guarantee exists for any other reason, Diehl AKO can request a lump-sum for processing from the Warranted User for each product delivered and in addition demand the costs for its return transport to the Warranted User.

All claims from this manufacturer's guarantee (included but not limited to the Guarantee Claims) shall be barred 6 months after occurrence of the fault, but not later than 3 months after the expiry of the Warranty Period.

8. Exclusion reasons

All Guarantee Claims of the Warranted User have been ruled out to the extent that and as soon as one or more of the following cases exists, occurs or has occurred:

- ➔ improper use
- ➔ assembly not done properly or correctly or according to standards or according to the installation instructions or information from Diehl AKO (including the installation and operating instructions for the string inverters for photovoltaic systems of the PLATINUM series in question)
- ➔ operation or use not carried out properly or correctly or according to standards or according to the installation instructions or information from Diehl AKO (including the installation and operating instructions for the string inverters for photovoltaic systems of the PLATINUM series in question)
- ➔ operation with or using defective protection devices
- ➔ unauthorised changes or repairs of any kind
- ➔ use of replacement parts and accessories not corresponding with the original specifications of Diehl AKO
- ➔ failure to perform ongoing maintenance in accordance with the maintenance instructions or information from Diehl AKO (including the installation and operating instructions for the string inverters for photovoltaic systems of the PLATINUM series in question)
- ➔ removal, damage or destruction of the sealing or the name plate attached by Diehl AKO
- ➔ effects of foreign bodies and force majeure

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- ➔ failure to comply with the relevant safety directives for the installation or the operation of the string inverters for photovoltaic systems of the PLATINUM series in question
- ➔ transport damage
- ➔ lightning strikes into the PLATINUM inverter, the photovoltaic power system in question or into the voltage supply mains to which the inverter has been connected, including in each case excess voltage resulting therefrom.

9. Assignability of the guaranty

This guaranty agreement including the Guarantee Claims resulting therefrom can only be assigned to a third party by a Warranted User with prior written consent of Diehl AKO. This guaranty shall automatically expire in the event of removal of the Warranted Product from the original place of installation and operation and re-positioning at a different location.

As a deviation from the above provisions, this guarantee agreement including the Guarantee Claims resulting therefrom can be assigned by the Warranted User to a third party operator without the express consent of Diehl AKO if (i) said third party operator acquires the Warranted Products from the Warranted User, (ii) said acquisition is proven to Diehl AKO in writing, naming the third party operator, (iii) the installed Warranted Product remain unchanged and (iv) the third party operator declares its agreement to these guarantee terms to Diehl AKO in writing.

10. General provisions

All claims of the Warranted User from this manufacturer's guarantee can only be assigned to third parties with the prior written consent of Diehl AKO.

If a provision of this manufacturer's guarantee is or becomes ineffective, the validity of the remaining provisions of the manufacturer's guarantee shall remain unaffected. The provision which is or becomes ineffective shall automatically be deemed replaced by a provision coming as close as possible to the provision which is or becomes ineffective. The above regulation shall apply accordingly in the event of a loophole.

This manufacturer's guarantee shall exclusively be governed by the law of the Federal Republic of Germany, ruling out the regulations of international private law (conflict of laws) and the UN convention concerning the international sale of goods.

The exclusive place of jurisdiction for all disputes out of or in connection with this manufacturer's guarantee shall be Wangen im Allgäu, Federal Republic of Germany, to the extent that it is a question of an Warranted User which is (i) a merchant or (ii) a private person without a general place of jurisdiction within the Federal Republic of Germany. Otherwise, the places of jurisdiction according to the Code of Civil Proceedings shall remain in effect.

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